MechanicaFluidorum

A Verifier-in-the-Loop Program Motivated by 3D Navier–Stokes Regularity

Dual-Scale Geometry, the Symmetric-Square Lock, and
the Truncated Viscous Katz–Pavlović Dyadic Shell Model

Scientific Progress Report — Intermediate Results,
Revised After the External Audit of 13 August 2026

Xavier Callens
SocrateAI Scientific Programme
`SocrateAI-Scientific-MechanicaFluidorum`

13 August 2026

Repository state: commit `a35cb92`+ | All verification gates PASS (exit 0)

Abstract

This report documents the state of a staged, verifier-in-the-loop research programme on the global regularity of the three-dimensional incompressible Navier–Stokes equations (NSE), organised around *Hypothesis U* — a uniform-in-cutoff enstrophy bound for the frequency-truncated system. The programme's distinguishing feature is not a claimed proof but a *method*: every assertion carries an explicit epistemic tier, mechanically enforced by a two-gate verification pipeline (exact rational arithmetic; Lean 4 kernel compilation with a pinned axiom footprint). We report 72 theorems kernel-compiled across seven Lean files — of which 65 carry Tier A claims, the remaining 7 having been *demoted* to Tier C by the audit below — six exact-arithmetic certification harnesses, and, equally importantly, a documented record of what has *failed*: two abandoned numerical designs, five caught errata (one in a formula sourced from the literature, one in the programme's own design memo, one a broken commit, one a false positive in the verification gate itself, and one an unpinned dependency that would have verified the programme against the wrong prover library), four analytical tracks that remain correctly blocked on undefined objects, and — the largest single failure in the programme's history — **an external expert audit that killed this report's own central framing claim** (§2). **The retraction is stated in full and accepted without qualification**: the formalisation, with a one-dimensional index cutoff, does not restate the unreduced three-dimensional problem; it describes a dyadic shell hierarchy. The programme has been retargeted accordingly, and its formal object is now the truncated viscous Katz–Pavlović dyadic shell model. **No claim is made that Hypothesis U is true, in any form, nor that the Navier–Stokes problem is resolved.** The Millennium Prize problem remains open and is not addressed by anything proven here.

Contents

1 Context Management: Reading This Document and This Repository **3**
1.1 The governing documents and their precedence 3
1.2 The epistemic charter 3
1.3 The three-tier gating system 4
1.4 The honesty clause and the obstruction ledger 4
1.5 Reproducing the verification state 5

2 The Problem, Stated Honestly **5**
2.1 Classical setting . . . 5
2.2 The regularised family and Hypothesis U . . . 6

3 Dual-Scale Geometry and T-Duality **7**
3.1 Physical motivation (**Tier C**) and its strict boundary . . . 7
3.2 The effective radius: what is actually proven (**Tier A**) . . . 7
3.3 The Symmetric-Square Lock (**Tier A**) . . . 8

4 The "Holographic Algebra" Organising Concept **9**
4.1 The picture . . . 9
4.2 Why this picture would matter, *if* it could be substantiated . . . 9

5 The Dyadic Shell Model: Established Results **9**
5.1 Energy flux telescoping and monotonicity (**Tier A**) . . . 10
5.2 The exact enstrophy-production identity (**Tier A**) . . . 10
5.3 A local production bound (**Tier A**) . . . 11

6 Statement-Level Formalisation, the Pivot, and the Demotion **11**
6.1 The shell-model Hypothesis U as a formal statement (**Tier A**, DRAFT) . . . 11
6.2 Closing verdict A5: the nonlinearity is now concrete (**Tier A**) . . . 11
6.3 The conditional Millennium reduction (**Tier C**, DEMOTED, DRAFT) . . . 12

7 The Four Analytical Tracks: Plan and Blocking Status **13**
7.1 Current status: correctly blocked . . . 15
7.2 Executable instrumentation delivered (**Tier B**) . . . 16

8 Numerical Investigation and Its Two Failures **16**
8.1 Resolution and the current evidence base (**Tier C** steering data) . . . 16

9 The 3-D Bridge: Instantiating the True Nonlinearity (OP-6) **18**
9.1 From computational check to kernel proof (**Tier A**) . . . 19
9.2 Repairing the reduction: giving the hypotheses content (**Tier C**) . . . 21
9.3 Eliminating a duplication hazard at the root . . . 21

10 Three Proposed Mechanisms, Killed by Measurement **21**

11 The Fourth Translation: Planar Confinement, and the Torus Theorem **22**
11.1 The attractivity experiment (σ), and a trap caught by its null model . . . 23
11.2 The neighbourhood theorem, scoped by a full read . . . 24
11.3 The torus theorem (**Tier A**) and the 5/6 ball conjecture . . . 24

12 The Live Band: Formalising Why $\alpha = 1/2$, and What Lies Below **24**
12.1 The screen, and a control broken by its own run . . . 25
12.2 Door #1, closed: there is no banded quartic invariant . . . 26

13 What Has Failed, and What That Taught **27**

14 Summary of Verified State **28**
14.1 What is genuinely established . . . 29
14.2 What is emphatically not established . . . 29

Note on reference verification **29**

References **29**

1 Context Management: Reading This Document and This Repository

Purpose of this section

This programme is executed collaboratively by a human mathematician and AI agents across many sessions. Context is therefore a *managed artifact*, not an accident of memory. This section specifies where authoritative context lives, how it is kept honest, and what a reader (human or machine) must load before acting. It is placed first deliberately: acting on this programme without this context has already caused documented failures (§13).

1.1 The governing documents and their precedence

Context is deliberately *not* held in any individual's or agent's memory. It is externalised into five documents with a strict precedence order:

Document	Role	Authority
`SPEC.md`	Normative rulebook	Highest. Defines the tier system, the honesty clause, the obstruction ledger. Where any other document conflicts with it, `SPEC.md` wins.
`PLAN.md`	Operational law	Task list with Definition-of-Done criteria and the escalation protocol (E-1…E-5). Wins over calendar/roadmap.
`LEDGER.md`	Claim inventory	**Normative:** a claim not listed here has no tier and may not be cited.
`LL.md`	Lessons learned	Why each process rule exists, with the incident that produced it.
`CLAUDE.md`	Build/verify manual	Commands, architecture, Mathlib location, working conventions.

1.2 The epistemic charter

The organising principle, stated in `SPEC.md` §0, is a division of labour:

The machine verifier checks the proofs; the human mathematician audits the questions.

Machine verification establishes that a proof is valid. It cannot establish that a *statement* means what the mathematics intends — that a formalisation is *adequate* to its informal gloss. Adequacy is reserved to human audit. Consequently **no LLM output gates a tier promotion**, and several artifacts in this repository are compiled, kernel-clean, and nonetheless explicitly marked DRAFT pending human statement-adequacy audit (§6).

1.3 The three-tier gating system

Tier	Meaning	Mechanically enforced gate
Tier C	Conjecture, analogy, physical narrative, unverified reduction	None, but must be tagged. Confined to `exploration/` or `docs/narrative/`. Floating-point code lives here and *only* here.
Tier B	Checkable: identities validated in exact rational arithmetic	Harness exits 0. `fractions.Fraction` or `int` only — floats are banned. **Every harness ships a negative control demonstrated to actually fail.**
Tier A	Established: Lean 4 kernel-compiled	Zero `sorry`; axiom footprint naming *no axiom outside* `{propext, Classical.choice, Quot.sound}`. A strict subset is cleaner and passes — see §13, where the earlier literal-equality reading is recorded as a defect.

Two subtleties are load-bearing and were learned the hard way:

1. **"No `sorry` in the source" is not the Tier A gate.** A failed proof still defines its theorem name in the environment; only `#print axioms` reveals a stray `sorryAx`. The gate is the *axiom footprint*, checked on the compiled artifact.
2. **A checker that cannot fail is not a checker.** Every Tier B harness must contain a negative control that is *demonstrated* to fail when the perturbation is applied. This report documents cases where an intended negative control turned out to be undetectable by construction, and had to be redesigned (§9).

1.4 The honesty clause and the obstruction ledger

Every public claim carries its tier. Tier A status attaches to the *formal statement actually proven*, never to its informal gloss. Additionally, `SPEC.md` §1.3 requires every proposed strategy — human or agent-generated — to file an *Obstruction Compliance Note* addressing five known barriers (Table 1). Strategies that cannot are rejected at review, before any formal work begins.

ID	Obstruction	Compliance question the strategy must answer
O1	Tao's averaged NSE blow-up	Which structural feature of the *true* nonlinearity, beyond the energy identity, does the strategy use?
O2	Supercriticality of enstrophy	Where exactly does the strategy gain over the naive enstrophy inequality, *uniformly* in α'?
O3	Caffarelli–Kohn–Nirenberg	If the strategy implies something stronger than CKN, what new mechanism pays for it?
O4	Buckmaster–Vicol non-uniqueness	In which class is the $\alpha' \to 0$ limit taken, and why is the limit the physical solution?
O5	The Euler test	Point to the step that fails at $\nu = 0$. A strategy that would also "prove" 3D Euler regularity is presumptively wrong.

Table 1: The obstruction ledger (`SPEC.md` §1.3). These are the barriers any credible attack must address; they are stated here because they constrain everything in §7. **Re-scoped 2026-08-13** by audit verdict D2, which found that merely *listing* these barriers against a one-dimensional sequence model overstates contact with them: O1 (Tao) and O5 (the Euler test) remain meaningful for the dyadic target; O2–O4 are inherited only if and when the 3-D bridge is built.

1.5 Reproducing the verification state

The only thing that counts as “it works” is the verification pipeline:

```
./scripts/verify.sh   # Gate 1  : Tier B exact-arithmetic harnesses
                      # Gate 1b : LEDGER consistency + single-active-file lint
                      # Gate 2  : Lean kernel compile + axiom-footprint membership test
```

Every dataset in `data/` ships a `.meta` sidecar recording the exact generating command, tool versions, and `sha256sum`, so results can be audited rather than merely re-read.

Gate 1b (added 2026-08-13). `LEDGER.md` is normative — a claim not listed there may not be cited — so an *orphan* row, one citing a theorem or file that no longer exists, silently degrades the whole inventory. Gate 1b checks that every repo path cited in the ledger exists, that every Lean name in a Tier A row is actually declared, and that no banned versioned duplicates (`_v2`, `_final`) have appeared. It carries its own negative controls, run on every invocation: a dangling path, a dangling name, and a banned filename must each be *rejected*, and a clean case must be *accepted*. It found one real defect on its first run (a row citing the non-identifier fragment `_half` rather than `step1_flux_bound_half`).

A latent reproducibility defect, found by doing the cold build

`lean_src/` was made a standalone Lake project on 2026-08-13, so Gate 2 no longer depends on an external Mathlib checkout this repository does not control. (That coupling was not hypothetical: an unrelated build running in that shared checkout rebuilt its `.olean` files mid-run and failed Gate 2 on a file this repository had not touched.)
Doing the build surfaced something more serious than the inconvenience that motivated it. The project’s `lakefile` declared its dependency as `require mathlib from git` with **no revision pinned**. A cold build from that specification resolves to Mathlib *master* — which in general does not build against the pinned toolchain, and, worse, would have silently verified every result against a *different Mathlib* than the one all existing `LEDGER.md` claims were checked with. For a programme whose entire epistemic claim rests on machine-checkability, an unpinned verifier is a defect in the foundation, not in the build script. Mathlib is now pinned by revision, and the lockfile pinning every transitive dependency is tracked rather than ignored.

2 The Problem, Stated Honestly

2.1 Classical setting

The incompressible Navier–Stokes equations on the torus $\mathbb{T}^3$ are

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u, \qquad \nabla \cdot u = 0, \qquad u(0) = u_0. \tag{1}$$

Leray established global existence of weak solutions [1]. The Clay Millennium Prize problem, as formalised by Fefferman [2], asks for a proof of *exactly one* of four statements — (A) existence and smoothness on $\mathbb{R}^3$; (B) existence and smoothness on $\mathbb{R}^3/\mathbb{Z}^3$ (the periodic case, which is the setting of this programme); (C) breakdown on $\mathbb{R}^3$; (D) breakdown on $\mathbb{R}^3/\mathbb{Z}^3$. Note that a *breakdown* result resolves the problem just as a regularity result does; this is consistent with the present programme’s stance that a negative verdict is a successful outcome. The obstruction is *supercriticality*: the natural energy bound is too weak to control the nonlinearity at small scales.

2.2 The regularised family and Hypothesis U

For $\alpha' > 0$ let $J_{\sqrt{\alpha'}}$ denote the projection onto Fourier modes $|k| \le 1/\sqrt{\alpha'}$, and let $u^{(\alpha')}$ solve the truncated system

$$\partial_t u + ((J_{\sqrt{\alpha'}} u) \cdot \nabla) u = -\nabla p + \nu \Delta u, \qquad \nabla \cdot u = 0, \qquad u(0) = J_{\sqrt{\alpha'}} u_0, \tag{2}$$

which is globally smooth for each fixed $\alpha' > 0$.

Hypothesis 1 (Hypothesis U)**.** For data u_0 and horizon T,

$$\sup_{\alpha'>0} \sup_{0\le t\le T} \|\nabla u^{(\alpha')}(t)\|_{L^2(\mathbb{T}^3)} < \infty. \tag{3}$$

Proposition 2.1 (Millennium Reformulation; **Tier C** at paper level)**.** *If Hypothesis U holds for all smooth divergence-free u_0 and all T, then 3D Navier–Stokes is globally regular. The chain is: energy identity $\Rightarrow$ $L^\infty_t L^2_x \cap L^2_t H^1_x$ bounds $\Rightarrow$ Aubin–Lions compactness $\Rightarrow$ a strong-L^2 subsequential limit which is Leray–Hopf $\Rightarrow$ under U the limit lies in $L^\infty_t H^1_x \hookrightarrow L^\infty_t L^6_x$ $\Rightarrow$ the Serrin class $2/s + 3/q = 1/2 \le 1$ $\Rightarrow$ smooth [3].*

Retraction — please read before any other claim in this report

An earlier version of this report carried, in this box, the claim that "Hypothesis U is a restatement of the open core of the Millennium problem, not a reduction of its difficulty". That claim was submitted to external expert audit on 13 August 2026 and was killed (verdict D1). It is retracted here in full.
The retraction is not that the claim was too strong about *difficulty* — it is that the claim was false about *what had been formalised*. The audit found that the programme's Lean statements are indexed by a one-dimensional shell index $n \in \mathbb{N}$ rather than a frequency $k \in \mathbb{Z}^3$ (verdict A2: the index flattening destroys the lattice geometry, the density of states, and the triad constraint $k_1 + k_2 = k_3$); that the enstrophy weight was the exponential sequence 4^n rather than the Laplacian symbol $|k|^2$ (A3); and that the mode-interaction term B was an *unconstrained* parameter, so that certifying a bound for it was, in the auditor's words, "authenticating a mathematically empty envelope" (A5). Taken together, what the formalisation describes is a **dyadic shell hierarchy**, not the unreduced 3-D equations. The prose of (1)–(3) above was never wrong; the claim that the *formal artifacts* restated it was.
The new target. The owner accepted the verdict in full (`docs/Memo 1.md`) and retargeted the programme rather than defending the framing. The formal target is now **global regularity bounds for the truncated viscous Katz–Pavlović dyadic shell model** — not a proxy for the 3-D problem, but an object of study in its own right, and a respected active area of mathematical fluid mechanics (blow-up: Katz–Pavlović [7], Cheskidov [8]; regularity under strong dissipation: Barbato–Morandin–Romito [9]). A machine-verified regularity-or-blow-up theorem for that model would, to the authors' knowledge, be a first. The 3-D bridge — the $\mathbb{Z}^3$ reindexing, OP-2 and OP-6 (§9) — is explicit, open future work, and is never again to be an implicit claim of the formalisation.
Why this is recorded as a success of the method and not merely as a setback. This programme's Definition of Done (§14) does not include proving Hypothesis U. It includes obtaining human verdicts *whatever they are*. The tier system exists precisely so that a framing claim can be located, isolated, and killed without taking any kernel-verified theorem down with it — and that is what happened: every Lean file still compiles, and the retraction cost exactly the seven theorems that deserved to be demoted (§6). The full verdict table (A1–A5, B1–B5, C1, D1–D2) with dispositions is in `LEDGER.md`; the retraction text is normative in `SPEC.md` §1.2.

What survives the retraction, stated precisely

Hypothesis U (3) as an informal statement about the continuum truncated system, and Proposition 5.1 as paper-level mathematics, are unaffected — they were always **Tier C** at paper level and are so labelled above. What is retracted is the assertion that the *Lean formalisation* of §6 is a formalisation of *that* statement. Audit verdicts A1 (the quantifier order $\exists C\,\forall N$) and A4 (quantifying over every solution) were upheld, so the statement's *logical shape* was sound; it was its *referent* that was wrong. Under the pivot, A2 and A3 dissolve rather than being repaired: for the dyadic shell model the index *is* the object and the weight $4^n = k_n^2$ *is* the correct weight. A5 was repaired by instantiating B (§6).

3 Dual-Scale Geometry and T-Duality

3.1 Physical motivation (Tier C) and its strict boundary

The programme's geometric inspiration is T-duality in string theory. For a dimension compactified on a circle of radius R, the free-boson partition function is invariant under $R/\sqrt{\alpha'} \to \sqrt{\alpha'}/R$ with simultaneous exchange of momentum and winding quantum numbers — this is a genuine theorem about the CFT spectrum, established as Eq. (2.2.11) of [14]. The fixed point of that map is $R = \sqrt{\alpha'}$.

Boundary of the analogy — Tier C, quarantined

Reference [14] was retrieved and read for this report, and the distinction it forces is worth stating precisely, because the programme's earlier internal documents blurred it. The **rigorous** content is the exact $R \leftrightarrow \alpha'/R$ symmetry of the spectrum. The frequently quoted claim that T-duality entails a **"minimal length"** is a *physical interpretation* of that symmetry — the argument being that a string cannot distinguish R from α'/R, so sub-$\sqrt{\alpha'}$ geometry is not resolvable — and is **not itself a separately proved theorem** in the sources consulted.
This report therefore uses T-duality **only** as motivation for the *shape* of a cutoff function. The mechanism acting on the PDE is the Fourier cutoff $J_{\sqrt{\alpha'}}$ and nothing more. Deriving the truncated dynamics from a genuine metric deformation is an *open problem* of this programme (OP-1), not an assumption. All cosmological interpretation — bounce scenarios, dark-sector mass laws, K3/mock-modular numerology — is quarantined in `docs/narrative/`, may never be imported by `lean_src/`, and may never be cited in a tier promotion.

3.2 The effective radius: what is actually proven (Tier A)

Define, for $\alpha, R > 0$,

$$R_{\text{eff}}(\alpha, R) := \max\left(R,\ \frac{\alpha}{R}\right). \tag{4}$$

The following are kernel-verified in `lean_src/CallensDualScale.lean`, with axiom footprint exactly `[propext, Classical.choice, Quot.sound]`:

Formal name	Statement	Reading
`Reff_pos`	$0 < R_{\text{eff}}(\alpha, R)$	positivity
`Reff_ge_sqrt`	$\sqrt{\alpha} \le R_{\text{eff}}(\alpha, R)$	universal minimum scale
`Reff_bounce`	$R < \sqrt{\alpha} \Rightarrow R_{\text{eff}} = \alpha/R$	sub-cutoff scales reflect
`Reff_inertial`	$\sqrt{\alpha} \le R \Rightarrow R_{\text{eff}} = R$	macroscopic scales unmodified
`Reff_tdual`	$R_{\text{eff}}(\alpha, \alpha/R) = R_{\text{eff}}(\alpha, R)$	self-dual symmetry
`Reff_gt_sqrt_of_ne`	$R \ne \sqrt{\alpha} \Rightarrow R_{\text{eff}} > \sqrt{\alpha}$	strictness
`Reff_eq_sqrt_iff`	$R_{\text{eff}} = \sqrt{\alpha} \iff R = \sqrt{\alpha}$	**sharpness**: the minimum is attained at exactly one point

Scope — what these theorems are and are not

These are theorems about the real-valued function $\max(R, \alpha/R)$. They are elementary, they are correct, and they are *not yet about fluids*. Their role is to fix the geometric language precisely; the parameter α' is carried as a *hypothesis parameter* $(0 < \alpha)$, never as an axiom, because Hypothesis U itself quantifies over it. A prior version of this repository asserted these as proven when no such proofs existed; that defect and its correction are recorded in `LEDGER.md` under “Retired / corrected claims”.

3.3 The Symmetric-Square Lock (Tier A)

The programme’s second structural ingredient is an algebraic rigidity constraint. If a sequence satisfies a second-order recurrence with operator L_2,

$$u_{n+2} = a\,u_{n+1} + b\,u_n,$$

then $v_n := u_n^2$ satisfies the third-order recurrence

$$v_{n+3} = (a^2 + b)\,v_{n+2} + b(a^2 + b)\,v_{n+1} - b^3\,v_n, \tag{5}$$

which is the symmetric square $L_3 = \mathrm{Sym}^2(L_2)$. If L_2 has characteristic roots $\{\lambda, \mu\}$, then L_3 has roots $\{\lambda^2, \lambda\mu, \mu^2\}$, and the coefficients of (5) are precisely their elementary symmetric functions:

$$e_1 = \lambda^2 + \lambda\mu + \mu^2 = a^2 + b, \qquad e_2 = -b(a^2 + b), \qquad e_3 = (\lambda\mu)^3 = -b^3.$$

This spectral content is verified as `sym2_symmetric_functions` — it is what makes the identity *structural* rather than a coincidence. The symmetric square of a second-order operator being third-order is classical in the theory of linear differential operators and Picard–Fuchs equations [15].

Intended role (Tier C). The *conjectural* purpose of the lock is to tie macroscopic transport (L_3) rigidly to a microscopic fiber operator (L_2), so that high-frequency modes cannot evolve as independent degrees of freedom — the hoped-for mechanism against Tao-type “choreographed” blow-up [4]. **Its relevance to Navier–Stokes is a conjecture of this programme and is not established.** It must be earned in the dyadic laboratory (§5).

4 The "Holographic Algebra" Organising Concept

Tier and status

The name *HoloAlg* (Holographic Algebra) denotes the programme's organising picture, not a proven theory. **Everything in this section is Tier C** except where it explicitly cites a Tier A result from §3. It is included because a reader must understand the motivating picture to follow why the four tracks (§7) were chosen — not because the picture carries evidential weight.

4.1 The picture

The organising idea combines two elements already made precise:

1. **A dual-scale geometry** in which the effective scale seen by the fluid is $R_{\text{eff}}(\alpha', R) = \max(R, \alpha'/R)$, so that (i) macroscopic scales are unmodified (`Reff_inertial`: "inertial invisibility"), and (ii) no scale below $\sqrt{\alpha'}$ is ever resolved (`Reff_ge_sqrt`). The regularisation is thereby confined strictly to the sub-cutoff regime.

2. **An algebraic lock** (§3) constraining macroscopic modes to be generated by products of microscopic ones, restricting the admissible spectrum to $\{\lambda^2, \lambda\mu, \mu^2\}$.

The label "holographic" is used in the loose sense that macroscopic (bulk) transport is determined by — and rigidly constrained by — data on a lower-dimensional algebraic structure (the fiber operator's spectrum). **No claim is made of a relationship to the AdS/CFT correspondence or any holographic duality in the technical sense of theoretical physics.** No such correspondence has been constructed here, and none is used.

4.2 Why this picture would matter, *if* it could be substantiated

For fluid mechanics, the mechanism would matter because it addresses obstruction O1 (Table 1) with a *structural* feature rather than an energetic one: Tao's averaged system satisfies the energy identity yet blows up, so any successful strategy must use something beyond energy. Algebraic rigidity of the mode spectrum is such a candidate. For astrophysics and cosmology, the same cutoff geometry has been *speculatively* associated with bounce scenarios — but that association is quarantined narrative, carries no evidential weight here, and is deliberately excluded from the mathematical core.

The programme's own structural finding against itself

When the four tracks' missing definitions were drafted (`docs/designs/TRACK_DEFINITIONS_DRAFT.md`), the drafting process surfaced a negative structural result about the programme's own architecture: under the natural choice of entropy functional, **Track T3 collapses into the direct production-identity attack rather than being independent**, and **T1 and T4 both depend on the same unproven embedding (OP-2)**, so agreement between them would be correlated rather than independent evidence. The advertised "four independent attacks" is therefore *not* currently four independent attacks. This finding is recorded rather than suppressed.

5 The Dyadic Shell Model: Established Results

This section was written when the truncated dyadic shell model was the programme's *laboratory* — a place to earn techniques before touching the continuum PDE. After the audit of 13 August

2026 it is the programme's *target* (§2). The results below are unchanged by that reclassification; what changed is that they are no longer preliminaries to something else. The model is due to Katz and Pavlović [7], whose published form (verified by retrieval, as restated in [8]) is $\frac{d}{dt}u_n + \nu 2^{2\alpha n}u_n - \lambda^n u_{n-1}^2 + \lambda^{n+1}u_n u_{n+1} = g_n$. The system used here is the unforced case with λ^n playing the role of k_n: shells $n = 0, \ldots, N$ carry wavenumbers $k_n = 2^n$ and real amplitudes a_n, with

$$\frac{da_n}{dt} = \mathrm{NL}_n - \nu k_n^2 a_n, \qquad \mathrm{NL}_n := k_{n-1}a_{n-1}^2 - k_n a_n a_{n+1}, \tag{6}$$

with truncation conventions $a_{-1} = 0$, $a_{N+1} = 0$. Energy is $E = \frac{1}{2}\sum a_n^2$ and enstrophy $\Omega = \frac{1}{2}\sum k_n^2 a_n^2$.

Scope note (honesty clause)

This model is **not** the Navier–Stokes equations. Its nonlinearity is **never** asserted in this repository to be derived from, or to approximate, $(u \cdot \nabla)u$; the audit of 13 August 2026 (§2) exists precisely because an earlier framing let that boundary blur. Results below are exact statements about finite sums and their consequences; no existence theory for the ODE system is invoked. What the model *is* is a well-studied object with a real literature and real open questions — which is what makes it a legitimate target rather than a proxy.

5.1 Energy flux telescoping and monotonicity (Tier A)

In `lean_src/DyadicShells.lean`: the nonlinear energy flux telescopes for *arbitrary* k and a (`dyadic_flux_telescopes`); it vanishes under the truncation $a_{N+1} = 0$ (`dyadic_flux_zero_of_boundary`); and consequently $dE/dt \le 0$ for $\nu \ge 0$ (`energyRate_nonpos`). Non-vacuity is witnessed: a concrete instance pins $dE/dt = -1$ exactly, so the inequality is not the trivial $0 \le 0$.

5.2 The exact enstrophy-production identity (Tier A)

The programme's most substantive dyadic result concerns *enstrophy*, the quantity Hypothesis U is about. Unlike energy flux, the enstrophy production sum does *not* telescope for arbitrary k; it collapses only under the doubling hypothesis $k_{n+1} = 2k_n$, and it collapses not to zero but to a multiple:

Theorem 5.1 (Dyadic enstrophy production; `enstrophy_production_dyadic`). *Under $k_{n+1} = 2k_n$ and truncation $a_{N+1} = 0$,*

$$\sum_{n=0}^{N} k_n^2 a_n \mathrm{NL}_n \;=\; 3 \sum_{n<N} k_n^3 a_n^2 a_{n+1}, \tag{7}$$

so that the enstrophy evolves as

$$\frac{d\Omega}{dt} \;=\; 3\sum_n k_n^3 a_n^2 a_{n+1} \;-\; \nu \sum_n k_n^4 a_n^2. \tag{8}$$

This is the dyadic analogue of vortex stretching, made exact: a signed production term competing against dissipation. The coefficient 3 is not fitted — it is $r^2 - 1$ for doubling ratio $r = 2$, and the Tier B harness independently confirms the general formula at $r = 3$ (coefficient 8). A kernel-verified negative control (`negative_control_nondoubling`) establishes that the identity genuinely *fails* for non-doubling k; the checker can fail.

5.3 A local production bound (Tier A)

Writing $S_N := \sum_{n<N} k_n^3 a_n^2 a_{n+1}$ and $\Omega_N := \frac{1}{2}\sum_{n\le N} k_n^2 a_n^2$, `lean_src/EnstrophyProductionBound.lean` proves, in a sqrt-free three-step chain (elementary algebra, $\sum x^2 \le (\sum x)^2$ for nonnegative x, Cauchy–Schwarz):

$$S_N^2 \;\le\; 2\,\Omega_N^3. \tag{9}$$

Precise scope of (9) — this is where over-reading would be easy

This is a *local, instantaneous, algebraic* bound. It does *not* use dissipation, it does *not* integrate in time, and it does *not* address uniformity in N. Note the exponent: production is bounded by $\Omega^{3/2}$, which is exactly the supercritical scaling that makes the enstrophy differential inequality $\dot{\Omega} \lesssim \Omega^{3/2}$ insufficient for global control. **Inequality (9) is therefore consistent with blow-up and is not evidence for Hypothesis U.** It is infrastructure: an exact, reusable, verified inequality, nothing more. The Tier B mirror further found, honestly, that the negative control breaks Step 1 under non-doubling k but does *not* break the final bound on the tested state family — an unforced negative result, recorded.

6 Statement-Level Formalisation, the Pivot, and the Demotion

6.1 The shell-model Hypothesis U as a formal statement (Tier A, draft)

`lean_src/DyadicShell_Statements.lean` (renamed 2026-08-13 from `HypothesisU_Statements.lean`, Memo Task 1) gives a non-vacuous Galerkin-shape formalisation. Writing $u : \mathbb{R} \to (\mathbb{N} \to \mathbb{R})$ for mode amplitudes, a solution at cutoff N satisfies: (i) $u(0) = \text{truncate}\, N\, u_0$; (ii) for every t and every retained mode $n \le N$, $\frac{d}{ds}u(s,n)|_{s=t} = B_n(u(t)) - \nu w_n u(t,n)$; (iii) $u(t,n) = 0$ for $n > N$. Then

$$\text{HypothesisU} \;:\equiv\; \exists C > 0.\ \forall N.\ \forall u.\ [u \text{ solves at cutoff } N] \Rightarrow \forall t \in [0,T].\ \Omega_N(u,t) \le C. \tag{10}$$

Why the quantifier order is the entire content

$\exists C\,\forall N$ — the constant is chosen *before* the cutoff. Swapping to $\forall N\,\exists C$ would be the trivial statement that each finite-dimensional truncation is bounded on its own, which carries no information about the $\alpha' \to 0$ limit. That swap is the failure mode this ordering exists to exclude.

Two prior defects were repaired by explicit human decision (recorded as Q1, Q2): initial data must be *projected* per cutoff, else the hypothesis set is empty at low N and the statement holds vacuously; and the weight is fixed concretely to $w_n = (2^n)^2$, so that "enstrophy" is provably a nonnegative sum of squares rather than merely a name. The file also *machine-checks the refutation* of the prior formalisation: quantifying over all smooth fields, without the equation as a constraint, yields a statement that is **provably false** (`unconstrained_bound_false`), since scaling a field by c scales enstrophy by c^2.

6.2 Closing verdict A5: the nonlinearity is now concrete (Tier A)

Until 13 August 2026 the mode-interaction term B in (10) was an **abstract parameter**, and the previous version of this report named that the decisive open item. The audit put it far more sharply (verdict A5): certifying a bound for an unconstrained B is a "fatal flaw ... authenticating a mathematically empty envelope" — an arbitrary truncated sequence system simply blows up, so the envelope contains nothing. Under the pivot this was repaired directly, by instantiating B as the Katz–Pavlović shell nonlinearity itself (Memo Task 2):

$$\texttt{shellB}: \quad B_0(v) = -(k_0 v_0 v_1), \qquad B_{m+1}(v) = k_m v_m^2 - k_{m+1} v_{m+1} v_{m+2}. \tag{11}$$

The bottom shell has no influx, so k_{-1} is never evaluated; for $n \geq 1$ this is exactly NL_n of (6). The structural property whose absence the audit called fatal is now a kernel-checked theorem, obtained from an exact telescoping identity:

Theorem 6.1 (Shell energy pairing; `sum_mul_shellB`, `shellB_energy_conservation`)**.** *For all $k, v : \mathbb{N} \to \mathbb{R}$ and all N, writing $\mathrm{out}_m := k_m v_m^2 v_{m+1}$, each summand satisfies $v_n B_n(v) = \mathrm{out}_{n-1} - \mathrm{out}_n$ (reading $\mathrm{out}_{-1} = 0$), so that*

$$\sum_{n \leq N} v_n\, B_n(v) \;=\; -\,k_N v_N^2 v_{N+1}, \qquad \textit{and hence} \qquad v_{N+1} = 0 \implies \sum_{n \leq N} v_n\, B_n(v) = 0.$$

The truncation hypothesis $v_{N+1} = 0$ is clause (iii) of `IsGalerkinSolution`, so this transfers to solutions with no extra assumption: `galerkin_shellB_conservation` states that every Galerkin solution of the concrete model conserves energy exactly, at every time. The viscous term of `IsGalerkinSolution` is $-\nu w_n u_n = -\nu k_n^2 u_n$, which is already the shell model's dissipation verbatim; together with (11) this makes

$$\texttt{DyadicShellHypothesisU} \;:\equiv\; \texttt{HypothesisU}\ \nu\ T\ (\texttt{shellB}\ k)\ (n \mapsto k_n^2)\ u_0$$

the programme's headline statement, in which **nothing abstract remains**. It is a statement about precisely the truncated viscous Katz–Pavlović system — the same model whose exact enstrophy-production identity is proven in §5.

What this does and does not fix

It fixes A5, and only A5. `DyadicShellHypothesisU` is a *statement*; it is not proven, and nothing here is evidence that it is true. The abstract-B scaffolding is retained in the file solely because the refutation theorems and the inhabitation witnesses quantify over it; it carries no headline claim. And instantiating B with the *true 3-D* nonlinearity remains untouched and open — that is OP-6 (§9), now correctly labelled as the 3-D bridge rather than as this file's missing piece.

6.3 The conditional Millennium reduction (Tier C, demoted, draft)

`lean_src/MillenniumReduction.lean` encodes the logical shape of Proposition 5.1 with the two undischarged analytic steps as *named hypothesis parameters*, never axioms:

$$\texttt{millennium\_reduction} : (\forall T,\ \text{HypothesisU}) \to \text{AubinLionsStatement} \to \text{ProdiSerrinStatement} \to \text{GlobalRegu}$$

The proof is the one-line composition `hPS (hAL hU)`. The intent was that all mathematical weight be parked in the hypotheses' *types*, so the Lean kernel would track the outstanding debt in the type signature rather than in the axiom environment — the pattern prescribed after a prior version of this work carried a raw `axiom aubin_lions_compactness`.

An adversarial self-review against `SPEC.md`'s actual text found three statement-adequacy gaps, all repaired by explicit decision (Q3–Q5): global regularity had covered only *time*-regularity, and now additionally requires spatial smoothness via membership in every Fourier-side Sobolev class H^s; the theorem had fixed a single horizon T while concluding a T-independent result, and now quantifies over all T; and the weight nonnegativity $w \geq 0$ is now an explicit hypothesis.

Demotion to — audit verdict B1

That intent did not survive audit. `AubinLionsStatement` and `ProdiSerrinStatement` are bare `Prop` parameters, and the auditor's finding (B1) is that bare Prop → Prop arrows "completely bypass PDE theory. The Lean kernel is merely verifying a logical tautology ($A \to B \to C$)" — which violates the Definition of Done's requirement of bona fide statements. **The verdict is accepted and the demotion is executed: all 7 theorems in this file are Tier C as of 2026-08-13.** They remain kernel-compiled and wired into Gate 2 so that they cannot silently rot, but they may no longer be cited as Tier A.

The distinction being drawn is sharp and worth stating plainly: parking analytic debt in a hypothesis's *type* is only honest if that type has content. Here it had none — a Prop variable constrains nothing — so the type signature recorded the debt in name only. The repair is Memo Task 4: import or define the required sequence spaces (ℓ^2, ℓ^p), state the Aubin–Lions and Prodi–Serrin analogues *for the shell model* with their actual topological content, and (audit item B4) specialise the reduction to the concrete object rather than to a generic B. That is the next design task and is **open**; `PLAN.md` §10 records that no implementation may begin before the design memo exists.

One audit item was repairable immediately (B2, revise). A previous independent review had raised, and this report previously recorded as an open candidate issue, whether the untruncated solution may differ per horizon T without a coherence clause. The audit confirmed it: the existential had to be hoisted. It now is — `HasGlobalBoundedLimit` fixes *one* limit trajectory $u_{\lim}$ across *all* horizons T, rather than permitting a different limit for each. The Cantor diagonal argument that would actually produce such a trajectory is thereby relocated into `AubinLionsStatement`'s type, where the audit said it belongs, instead of being silently absent; formalising the diagonalisation itself is part of the Task 4 repair.

Audit status of the two files after the pivot

`DyadicShell_Statements.lean` is **Tier A** and DRAFT: it compiles clean with correct axiom footprints, its quantifier shape was upheld (A1, A4), its nonlinearity is now concrete (A5), but it has **not** passed a full human statement-adequacy audit. `MillenniumReduction.lean` is **Tier C** and DRAFT pending the Task 4 repair. The general lesson is the one this section now demonstrates twice over: fixing the gaps found in a self-review (Q3–Q5) did *not* certify that no others remained, and the ones that remained were the structural ones. Authorship — including adversarial self-review — never licenses a claim. Only human audit does, and when it was finally obtained it removed a claim rather than confirming one.

7 The Four Analytical Tracks: Plan and Blocking Status

Stage 3 of the programme proposes to prove Hypothesis U *scale-by-scale* rather than as a monolith, partitioning Fourier space into dyadic shells and controlling $\sum_j E_j^{(\alpha')}(t)$. Four deep analytical disciplines are proposed as independent attacks. Each is an *analogy with a falsification plan*, and each carries a kill criterion agreed in advance.

Status of this section after the pivot

All four tracks are formulated against the *continuum* problem on $\mathbb{T}^3$ and depend on $\mathbb{Z}^3$ lattice structure (T1's triads, T4's percolation field). They therefore sit **downstream of the 3-D bridge** (§9), not of the programme's current formal target. They are retained here unchanged, and unchanged also in their status: blocked. Their blockage was never contingent on the framing that was retracted — it is contingent on four objects that do not exist, listed below.

	Discipline	Proposed mechanism	First falsifiable milestone / kill criterion
T1	Bourgain–Demeter ℓ^2 decoupling [10]	Arithmetic depletion: the Sym^2 constraint starves triadic resonances $k_1 + k_2 = k_3$	Exact count of constrained vs. unconstrained resonant triads. **Kill:** constrained count grows at the same order $\Rightarrow$ no depletion.
T2	Mouhot–Villani nonlinear Landau damping [11]	Phase mixing suppresses "enstrophy echoes" via a Newton-scheme argument in a high-regularity class	A precise *definition* of an enstrophy echo in the truncated system, plus its exact evolution. **Kill:** echoes not suppressed even in the dyadic model.
T3	Golse–Saint-Raymond hydrodynamic limits [12]	A relative-entropy functional controls dissipation uniformly in α'	A relative-entropy inequality for the truncated system at fixed α'. **Kill:** inequality already fails at fixed α'.
T4	Duminil-Copin percolation [13]	High-enstrophy regions form *subcritical* percolation clusters $\Rightarrow$ singular set of dimension 0	Recover the CKN dimension bound [5] in the lattice formulation. **Kill:** cannot recover even CKN.

A near-miss worth recording: the correction that would itself have been an error

An earlier draft of this report flagged an apparent defect in the programme's specification. The spec describes T2 as resting on *Gevrey-class* regularity; a first retrieval pass indicated that Mouhot–Villani's *core* result is set in *analytic* regularity, and the obvious move was to "fix" the spec by substituting one for the other.

That fix would have been wrong. Re-fetching the abstract directly — rather than relying on the intermediate summary the first pass produced, parts of which were flagged as paraphrase — gives *both*: the paper establishes "exponential Landau damping in **analytic** regularity", while the version note records that the main result "now covers Coulomb and Newton potentials, and (2) some classes of **Gevrey** data". The base theorem is analytic; the final version does cover Gevrey. **The spec was defensible as written and was left standing.**

The methodological point is the reason this is in the report at all: the first pass had enough material to justify editing the normative rulebook, and editing it would have *introduced* an error on the strength of a paraphrase — exactly the failure mode rule LL-6 exists to prevent, one level up from where it was first learned. **When a proposed correction rests on a summary rather than a primary quotation, flag it and re-fetch; do not edit the rulebook.**

One substantive caveat did survive, and it is the part that bears on feasibility: coverage is "*some* classes of Gevrey data", not all. Any T2 argument leaning on Gevrey regularity must identify *which* class and must not assume the full Gevrey scale.

A disanalogy that has not been resolved away

The Golse–Saint-Raymond programme [12] concerns the Boltzmann-to-Navier–Stokes hydrodynamic limit, whereas this programme's $\alpha' \to 0$ limit is **not** a Knudsen limit and has no kinetic layer — a disanalogy flagged internally (review finding M4) and repeated here so T3 is not over-read. This one stands.

7.1 Current status: correctly blocked

All four tracks are blocked on undefined objects — and this is the correct state

Tracks T1–T4 require four objects that **do not exist** in this programme or, to the authors' knowledge, in the literature:

- **OP-2**: the Sym^2-constrained spectrum on $\mathbb{T}^3$. The lock is proven for *scalar recurrences*; its embedding into 3D Fourier dynamics indexed by $k \in \mathbb{Z}^3$ is open. Any invented embedding would make T1's count meaningless.
- **OP-3**: "enstrophy echo" as a formula. Landau-damping echoes are defined in Vlasov phase space; no NSE analogue exists.
- **OP-4**: the entropy functional h and reference state $\bar{u}$. Choosing h *is* the mathematical content of T3.
- **OP-5**: the coupling of the percolation field to Sym^2. Without it, T4 measures generic percolation, which cannot beat CKN.

Under escalation rule E-1, an agent encountering a missing definition must **stop and escalate**, never improvise. Draft definitions have been authored and await human audit; *authorship never unblocks a track — only audit does.* Progress here is deliberately gated on human judgement rather than machine throughput.

7.2 Executable instrumentation delivered (Tier B)

Definition-independent tooling has nonetheless been built and certified, so that the tracks can execute immediately once unblocked: exact lattice representation counts $r_3(n)$ for $n \leq 10^4$ with Legendre anchors; exact unconstrained triad counts for $|k_i|^2 \leq M^2$; and an exact 3D periodic site-percolation toolkit (union-find over integers, wrap detection, 27 checks, negative control that fails when periodicity is dropped).

8 Numerical Investigation and Its Two Failures

The central empirical question is whether $\sup_t \Omega_N(t)$ grows with the cutoff N or saturates. Two successive designs *failed their feasibility premises*, for genuinely different reasons; both are documented rather than quietly replaced.

1. **Explicit float RK4 (D4).** Stiffness: stability requires $\Delta t \sim 1/(\nu k_N^2)$, which collapses as N grows. 21 of 36 configurations were computationally infeasible.
2. **Exact-rational IMEX-Euler (D5).** The implicit viscous treatment removed the stiffness barrier as designed — but the scheme failed anyway, for an unrelated reason: iterating a *nonlinear* rational map in exact arithmetic causes the numerators and denominators to grow geometrically in bit-length (hundreds of digits within 2–6 steps), while the state magnitude stays small. **This is a representation blow-up, not a numerical instability.** Implicit-versus-explicit was the wrong axis on which to redesign.

Generalised lesson, now a standing rule

Exact rational arithmetic is excellent for closed-form algebraic identities and inequalities proven in one shot, and poor for *iterating* a nonlinear map over many steps. For trajectories, use dual-precision floating point (Tier C steering) and reserve exactness for the algebra. After two failed designs, the methodology decision was escalated to the human owner rather than a third variant being chosen unilaterally.

8.1 Resolution and the current evidence base (Tier C steering data)

The accepted resolution was to stop pursuing exact-rational certification and retain dual-precision (float64 + 50-digit `mpmath`) runs as permanent Tier C steering data. Across the full grid $N \in \{8, 12, 16, 20, 24\}$, several low-viscosity configurations initially reported divergence, with the two precisions agreeing to $\sim 10^{-14}$ — ruling out rounding, but *not* distinguishing genuine divergence from an under-resolved step size (the design's Δt formula never depended on ν). A step-size refinement study resolved this:

Finding	**Detail**
Divergences are artifacts	All four diverging $(\nu, \text{profile})$ pairs flip to `OK` at a finite refinement level and stay `OK` at every finer level, with $\sup \Omega$ *decreasing* as $\Delta t \to 0$ (e.g. $45.1 \to 21.8 \to 17.3 \to 15.8$) — the signature of a discretisation artifact resolving, not a fixed-time blow-up.
N-independence	44 of 45 configurations are *exactly* flat in N (bit-identical $\sup \Omega$ at every cutoff). **This was reported as striking evidence. It is not evidence at all — see the retraction immediately below.**
The one exception, explained	$\nu = 0.1$, profile P2 appeared to grow linearly in N $(4.5, 6.5, \ldots, 12.5)$. Traced by hand: this equals $\frac{1}{2}(N+1)$, which is profile P2's *analytic initial* enstrophy — P2 excites every retained mode by construction, so a larger cutoff starts with more enstrophy independent of any dynamics. The trajectory collapses immediately from $t = 0$ and decays thereafter. **An initial-data artifact, not dynamical growth.**

Retraction (14 August 2026): the N-flatness is an artifact of the grid

An earlier version of this report presented the flatness above as the evidence base for a uniformity verdict. **That reading was wrong, and the error was the report's own.** Direct measurement of the shell-amplitude profile shows the cascade never reaches the shells the grid varies. At $\nu = 10^{-3}$ the enstrophy is carried almost entirely by a *single* shell:

shell	peak $\lvert a_n \rvert$	$k_n^2 a_n^2$	share of $2\Omega \approx 140$
$n = 8$	4.5×10^{-2}	132.7	$\approx 95\%$
$n = 10$	1.7×10^{-6}	3.0×10^{-6}	2×10^{-8}

The grid's *smallest* cutoff was $N = 8$ — exactly one shell above the decisive one. Raising N from 8 to 24 appended shells contributing $\sim 10^{-8}$ relative, which is why 44 of 45 configurations agreed *bit-identically*: adding numerically zero shells changes $\sup \Omega$ by nothing. **The experiment varied the one parameter along the one range in which it could have no effect.** Lowering to $N = 7$ would have removed 95% of the enstrophy.

"Flat in N" is therefore close to a tautology here and cannot distinguish *"the bound is uniform in the cutoff"* from *"no regime was tested in which the cutoff binds"*. No verdict may be issued on this evidence base, and none has been.

The failure mode is now mechanically excluded

A Tier B harness (`tests/tier_b_grid_adequacy.py`, wired into Gate 1) rejects such a grid *before it runs*, on exact integer arithmetic. Balancing viscous against nonlinear rates with the K41 amplitude $a_n \sim k_n^{-1/3}$ gives $k_d \sim \nu^{-3/4}$; with $k_n = 2^n$ the cutoff lies at or above the dissipation scale iff $2^{4N}\nu^3 \geq 1$, i.e. writing $\nu = p/q$, iff the integer inequality $2^{4N}p^3 \geq q^3$ holds. No floats, no logarithms.

Its negative control is the historical grid itself: **0 of 15 configurations had a biting cutoff**. The redesigned measurement (`docs/designs/D6_CUTOFF_BITING_MEASUREMENT.md`) sweeps N *across* the dissipation shell instead of above it, reports the production flux at the cutoff and the shell-population profile alongside $\sup \Omega$, and carries a $\nu = 0$ control where blow-up is a published theorem — the sharper control, because it is the one this programme's hoped-for conclusion would predict wrongly.

9 The 3-D Bridge: Instantiating the True Nonlinearity (OP-6)

This section describes work aimed at the *true* 3-D nonlinearity. It should now be read as work on the **bridge** between the programme's formal target (the shell model, §5, §6) and its motivating problem (§2) — open future work, and not a claim that the bridge exists. Note the distinction from §6: B in (10) is no longer abstract, having been instantiated by the shell nonlinearity (11); what remains open is instantiating it by the Fourier–Galerkin nonlinearity on $\mathbb{Z}^3$, which is a different and strictly harder object.

A repository-wide search established that **no Fourier-triad convolution formula exists anywhere in this programme's sanctioned content** — the triad tooling counts lattice points without amplitudes; the OP-2 draft self-admits it discards the angular structure of k "which is precisely where 3-D vortex-stretching geometry lives"; and the dyadic NL_n is never asserted to derive from $(u \cdot \nabla)u$. Instantiating B by the 3-D nonlinearity is therefore a genuine E-1 blocker.

On $\mathbb{T}^3$, with $u = \sum_k \hat{u}_k e^{ik\cdot x}$, $k \cdot \hat{u}_k = 0$, $\hat{u}_{-k} = \overline{\hat{u}_k}$, and Leray projector $P(k) = I - \frac{k \otimes k}{|k|^2}$, the truncated nonlinearity is

$$\partial_t \hat{u}_k = -\nu |k|^2 \hat{u}_k - i \sum_{\substack{p+q=k \\ p,q \in \Lambda}} P(k)\big[(q \cdot \hat{u}_p)\, \hat{u}_q\big]. \tag{12}$$

An erratum caught by exact arithmetic — a case study in why the method exists

The formula first recorded in the scoping memo, taken from a single literature cross-check, read $P(k)[q\,(\hat{u}_p \cdot \hat{u}_q)]$ — the *vector* q times the scalar velocity–velocity dot product. The Tier B harness then *proved*, in exact arithmetic, that this expression is **identically zero for every** k: $P(k)$ annihilates anything parallel to k, and since every pair (p, q) with $p+q = k$ appears alongside its swap (q, p), the two terms cancel exactly, $P(k)[qs] + P(k)[ps] = P(k)[(p + q)s] = P(k)[ks] = 0$. That cannot be the real nonlinearity. The corrected form (12) — q *dotted* with $\hat{u}_p$, a scalar, times the vector $\hat{u}_q$ — matches the Fourier transform of the convective derivative and is nonzero. The error was caught by computation, not by inspection.

The resulting harness (`tests/tier_b_nse_triad_convolution.py`, **Tier B**) certifies two structural facts in exact Gaussian-rational arithmetic at $M = 1, 2, 3$: *transversality* $k \cdot N(u)_k = 0$ (unconditional, since $P(k)$ projects onto $k^\perp$), and *detailed energy conservation* $\sum_k \overline{\hat{u}_k} \cdot N(u)_k = 0$ given divergence-free, conjugate-symmetric input. Two negative controls — dropping the Leray projection, and breaking the divergence-free condition on one mode — each break exactly one fact and leave the other intact, matching the hand derivation rather than merely "changing something".

Recorded honestly: several negative controls were undetectable by construction

Two natural "plausible bug" perturbations turned out to be invisible: swapping $p \leftrightarrow q$ throughout (the pair set is symmetric under relabelling for *every* fixed k, not merely after summing), and substituting the outer k for the summed q (the difference vanishes because $p \cdot \hat{u}_p = 0$ by divergence-freeness). The negative control actually used — multiplying by the wrong velocity factor — was found by *testing* candidates, not by assuming one would work. Additionally, the general triad identity underlying energy conservation is here verified *computationally* at three truncations, **not proven symbolically**; a Tier A proof remains future work.

9.1 From computational check to kernel proof (Tier A)

The harness above certifies two facts *computationally*, at three truncations, and its own docstring declines to claim either as proven. Both are now Tier A, in `lean_src/AbstractAlgebraicConservation.lean` — renamed 2026-08-13 from `TriadConservation.lean` under audit verdict C1, which found the old name OVERSTATES REACH: the file's abstraction bypasses geometric aliasing on $\mathbb{Z}^3$ entirely, and the name should say so. The docstring now states that exclusion explicitly:

Theorem 9.1 (Termwise triad cancellation; `triad_pairing`). *If $k_p + k_q + k_r = 0$ componentwise and the field is divergence-free at p ($k_p \cdot u_p = 0$), then the two orderings of the triad summand cancel* exactly*:*

$$(k_q \cdot u_p)(u_q \cdot u_r) \;+\; (k_r \cdot u_p)(u_r \cdot u_q) \;=\; 0.$$

Why the first attempt failed, and what actually worked

The harness records that a direct *three-way* relabelling (cycling $p \to q \to r$) did not close. It does not need to. The cancellation is **two-way**, and stronger than a summation argument: it is **termwise**. Pairing (p, q, r) with (p, r, q) — swapping only the last two indices, leaving p fixed — and using symmetry of the pairing followed by additivity,

$$[(k_q \cdot u_p) + (k_r \cdot u_p)](u_q \cdot u_r) = ((k_q + k_r) \cdot u_p)(u_q \cdot u_r) = (-k_p \cdot u_p)(u_q \cdot u_r) = 0.$$

This was re-derived by hand and confirmed on 6486 triples in exact arithmetic *before* any Lean was written. Summing it over any finite index set closed under the swap gives detailed energy conservation (`triad_sum_zero`); transversality (`transversality_of_sum`) is separate and unconditional, following from the projector landing in $k^\perp$.

Scope: this does not close OP-6

The file is stated over an *arbitrary commutative ring*, with an arbitrary additive index group and additive wavevector map. It is deliberately **not** stated over $\Lambda \subset \mathbb{Z}^3$ with complex velocities, because that concrete apparatus *is* the open decision D1. **The bridge from this lemma to the concrete Fourier–Galerkin setting is not built**; nothing here instantiates B; nothing here is a statement about Navier–Stokes solutions. One hypothesis is load-bearing rather than technical: the swap has fixed points (where $q = r$, forcing $p = -2q$) at which the pairing yields only $2f = 0$, so the ring must have no 2-torsion. For the intended instantiations ($\mathbb{R}$, $\mathbb{C}$) this holds.

Open decisions. Instantiating B is not a substitution but a restructuring: the state representation must move from $\mathbb{N} \to \mathbb{R}$ shell scalars to $\mathbb{Z}^3 \to \mathbb{C}^3$ Fourier vectors carrying divergence-free and reality constraints. Whether to undertake that full reindexing or seek a smaller shell-preserving proxy, and whether to sequence it before or after OP-2, are open questions reserved to the human owner.

A second, sharper limit on the pivot (14 August 2026)

The audit established that the formalisation describes a shell model rather than 3-D Navier–Stokes. A further fact bounds what the shell model can deliver even in principle.
The dyadic dissipation term is $\nu k_n^{2\alpha}$; the Navier–Stokes viscous term is $\nu \Delta u$, whose Fourier symbol is $-\nu|k|^2$. Hence $\alpha = 1$ *is* the Navier–Stokes dissipation — the same exponent, not an analogy. Cheskidov proves global regularity for $\alpha \geq 1/2$. Therefore:

The dyadic shell model at Navier–Stokes dissipation is provably regular.

The model's difficult regime, $\alpha < 1/2$, corresponds to dissipation *weaker* than Navier–Stokes. A result below $\alpha = 1$ is a result about *hypo-dissipative* shell models. That is live mathematics — Cheskidov singles out $\alpha = 1/3$ as carrying the same nonlinear estimates as 4-D Navier–Stokes — but it is **not a step toward 3-D Navier–Stokes and must never be reported as one**.

Correction (E-3b, 15 August 2026): the band is narrower, and there are two of them

Source verification commissioned before opening this track found this report's own statement of the band to be wrong. **Barbato–Morandin–Romito** (*Nonlinearity* **24** (2011) 3083–3097, Thm A) closed $\alpha \in [2/5, 1/2)$ in 2011 — with *uniqueness* and smoothness, from mere ℓ^2 data — for **positive** initial data. And *both* bounding theorems assume positivity (the blow-up theorem additionally needs *large* data), so there are two bands, not one: for positive data the open region is $[1/3, 2/5)$; for sign-changing data *nothing* is proven below $1/2$ in either direction. With intermittency dimension $d = 5 - 2/\alpha$, the residual positive-data band sits at $d \in [-1, 0)$, outside the physically relevant range — and the field's own survey (Cheskidov–Dai–Friedlander) states verbatim that BMR "settles that solutions to the dyadic model corresponding to the 3D NSE are globally regular". **The genuine room is sign-changing data.**
The failure mode is worth more than the correction. The Tier B gate encoding these thresholds had *both* controls, and both fired — but its positive-control anchor asserted that $\alpha = 2/5$ is open, which is precisely BMR's endpoint. A control tests the code against the premise you believe; it cannot test the belief. And LL-6 had already been applied to this very paper — to its *abstract*, which does not carry the β range. New rule (LL-16): a numerical threshold taken from the literature must be sourced to the *theorem statement* with its hypotheses, and a correction ships with a regression control pinning the value that was wrong.

An unplanned consistency check fell out of the same calculation. The K41 balance $\nu k^{2\alpha} \sim k\, a_k$ with $a_k \sim k^{-1/3}$ gives a dissipation scale $k_d \sim \nu^{-1/(2\alpha - 2/3)}$, whose exponent *diverges* as $\alpha \to 1/3^+$: below that threshold no dissipation scale exists, so the cascade is stopped at no finite shell. That is exactly Cheskidov's proven blow-up threshold, reached from an elementary heuristic that knows nothing of his proof.
Owner decision: this gap is to be treated as a gap to *close*, not a ceiling to accept. The programme will pursue the bridge — the $\mathbb{Z}^3$ embedding and the instantiation of the nonlinearity — both of which remain blocked on definitions that do not exist, and so admit only authored candidates submitted for human audit.

9.2 Repairing the reduction: giving the hypotheses content (Tier C)

Audit verdict B1 found the reduction's two hypothesis parameters to be bare Prop $\to$ Prop arrows: "the Lean kernel is merely verifying a logical tautology". The repair, carried out on 14 August 2026, is *not* to prove those steps — parking undischarged analysis in a named hypothesis parameter rather than an axiom is the pattern this programme deliberately adopted, so that the kernel tracks the debt in the type. The repair is to make the types say something.

What a compactness statement has to deliver

`IsGalerkinLimit` now carries four clauses, each load-bearing: the limit solves the untruncated system; it has *finite enstrophy* (a genuine ℓ^2 condition, since there is no cutoff to truncate the sum); that enstrophy is bounded on every horizon, with the constant quantified *inside* the horizon — compactness gives no uniformity in T, and writing $\exists C\,\forall T$ would claim more than the analysis provides; and the limit is genuinely the modewise limit of the Galerkin family **along a subsequence**. The subsequence is not a technicality: a compactness argument extracts one, and asserting convergence of the full sequence would overstate the mathematics.

The ℓ^2 upgrade was deliberately *localised*. Hypothesis U quantifies over *Galerkin* solutions, whose truncation clause forces $u_n = 0$ for $n > N$ — so the finite sum over $n \leq N$ already captures every nonzero mode and is *exact* there, not an approximation. Only the objects with no cutoff needed the infinite-series treatment. A bridge lemma (bounded partial sums of nonnegative terms $\iff$ summable with $\sum' \leq C$) makes the upgrade a conservative extension: nothing previously proven is invalidated.

The file remains Tier C, and two things remain unmachine-checkable

The repair makes the undischarged debt *legible*; it does not pay it. Both hypothesis parameters are still assumed. Moreover, whether the four clauses *say the right thing* is a statement-adequacy question reserved to human audit — it cannot be settled by the kernel, and that cost is scheduled rather than discovered later.

9.3 Eliminating a duplication hazard at the root

Until this point every file in `lean_src/` re-declared, verbatim, the definitions it needed from its neighbours, because the verification gate compiled each file in isolation and cross-file imports did not resolve. That convention carried a silent-drift hazard of exactly the kind this programme exists to eliminate: **a correction applied to one copy and not the other leaves both files compiling cleanly, both gates green, and the mathematics divergent**, with no mechanism able to see it.

With the project now self-contained, the gate builds the dependency graph first and then re-elaborates each file with the project's own build directory on the search path. Imports resolve; the duplication is gone; there is exactly one definition of each object in the repository. One implementation detail is worth recording because getting it wrong would be invisible: the gate must re-elaborate with `lean` rather than rely on `lake build` alone, because a *cached* build target emits no axiom output at all — which would make the footprint check silently vacuous rather than failing loudly.

10 Three Proposed Mechanisms, Killed by Measurement

On 14 August 2026 the programme's central conjectural mechanism — the Symmetric-Square lock — was tested in three successive formulations. All three failed, each at a cost of hours rather than months, and each for a variant of the same reason.

Proposed mechanism	**What it actually constrains**	**How it died**
Pointwise, $a_{2n} = c\,a_n^2$	the *amplitude*; the spectral slope stays free	On paper: for any power law $a_n = Ar^n$ the ratio is $1/A$, independent of r. Blow-up is characterised by the slope.
Recurrence signature, $c_2^3 + c_1^3 c_3 = 0$	genuinely discriminating (controls separate by 10^{14}) — but nothing specific to blow-up	By measurement: S rises from early to late in the blow-up regimes, *and just as much at $\alpha = 1$ where blow-up is provably impossible.*
Radial embedding on $\mathbb{Z}^3$, shell indices $\{2i, i+j, 2j\}$	the *parity* of the shell index	By counting, against a criterion fixed in advance: the permitted fraction *rises* with M ($0.019 \to 0.263 \to 0.525$), and at $M = 8$, 99% of permitted triads are same-shell.

The common defect, and why it is worth naming

Each formulation carried the *symbols* of the proven lock into a new setting while leaving free the quantity it was supposed to control. A translated constraint can look faithful, type-check, pass every gate, and constrain almost nothing. The cheap general test — now a standing rule — is to **compute the constraint's solution set in the new setting and check it is smaller than intended**: substitute the simplest family the setting supports and see whether the constraint restricts it.

The third death is the most instructive, because the criterion it failed (*"constrained count grows at the same order ⇒ no depletion ⇒ kill"*) had been written into the specification *before* the mechanism was proposed. The draft's own honesty clause had even anticipated the cause: a radial reduction discards the angular structure of k, which is where three-dimensional vortex stretching lives. Writing the kill criterion in advance is what turned a disappointment into a result.

What was built in their place

The resonant-triad hypergraph — vertices are Fourier modes, hyperedges are the resonances $k_1 + k_2 = k_3$ — is *definition-independent*: it is fixed by the Navier–Stokes nonlinearity itself and needs no blocked definition, so it could be built immediately. Its normalised-Laplacian spectral gap is stable at ≈ 0.8334 across $M = 2, 3, 4$, apparently approaching $\approx 5/6$ (suggestive on three points, not established).

The reading is a constraint on all future proposals: the triad structure is a **strong expander**, so no depletion is available for free, and any candidate lock must be shown to *destroy* that expansion. The vague question "does the locked sub-hypergraph have a spectral gap?" now has a concrete number to beat.

11 The Fourth Translation: Planar Confinement, and the Torus Theorem

The post-mortem of the three deaths of §10 produced a diagnosis rather than a fourth guess: all three translations searched for a *scarcity* mechanism (fewer resonances, smaller exponents) while the proven lock is a *closure* statement. Squaring a velocity field is the Sym^2 operation on its spectrum, $\mathrm{Sym}^2\{k_1, k_2\} = \{2k_1,\ k_1 + k_2,\ 2k_2\}$, and the closure of a mode pair under repeated triadic interaction stays in the 2-plane $\langle k_1, k_2 \rangle$ — computed exactly in this repository: six rounds, 2144 modes generated, zero outside the plane.

OP-2′: the lock's exact 3-D content is planar confinement

A flow whose spectrum lies in a 2-plane of $\mathbb{Z}^3$ is a **2D3C** flow (two-dimensional, three-component — the standard term; "2.5-D" also names a different class in parts of the literature). The 3-D equations then split into 2-D Navier–Stokes plus a passively advected scalar (Biferale–Buzzicotti–Linkmann, arXiv:1706.02371, quoted verbatim in the design memo), and 2-D regularity is classical (Ladyzhenskaya; attribution to Leray for global strong existence, per Ponce–Racke–Sideris–Titi). **K1, verified in exact arithmetic**: a field supported on the *tilted* plane $\langle (1, 0, 0), (0, 1, 2) \rangle$ has $N(u) = 0$ on all 322 out-of-plane modes under the Leray-projected truncated nonlinearity; the negative control (a generic field) leaks on all 322. The lattice/Galerkin form of this invariance was not found published as such; the continuum antecedents are Moffatt (J. Fluid Mech. 741 (2014) R3, cited by Biferale et al., itself unfetched) and Biferale et al. 2017. This candidate, unlike its three dead predecessors, was screened *before* authoring: the LL-11 vacuity test passes (plane sublattices are a measure-zero subfamily), and the depletion screen scores planes at $D = 1.87 > 1$ — the mechanism was never about triad counts.

11.1 The attractivity experiment (σ), and a trap caught by its null model

The honest research question OP-2′ produces is dynamical: *is the planar-locked manifold attractive, neutral, or repulsive under the truncated 3-D Galerkin flow?* The expected answer — repulsive, by the classical instability of 2-D flows in 3-D — is pre-registered as the null, with a theorem-backed positive control (exactly planar data must stay exactly planar; measured at literal 0.00 out-of-plane energy) and a negative control (generic data must show $O(1)$ out-of-plane fraction; measured 0.62–0.74). The pilot measured $\sigma = +0.57$ to $+1.04$: repulsive, ε-independent, faster at lower viscosity.

The full experiment ($M = 3$, coordinate *and* tilted planes, exact rational grid-adequacy stamps) produced one line that exercised the programme's discipline: at $\nu = 1/2$ the measured σ was *negative* (-0.265) — the pre-registered "candidate discovery" branch. Before interpretation, the obvious null model was computed in closed form: under pure linear decay $u_k(t) = u_k(0)e^{-\nu|k|^2 t}$, the out-of-plane fraction *already* drifts at $\sigma_{\mathrm{lin}} = -0.305$, because the out-of-plane modes carry a higher energy-weighted $\langle |k|^2 \rangle$ than the in-plane modes (6.26 vs 4.95). The measured excess $\sigma - \sigma_{\mathrm{lin}} = +0.04 > 0$: the nonlinearity pushes *out* of the plane even there, and the negative sign was spectral bookkeeping. The most seductive artifact is the one that would have announced a discovery; it was caught by the same rule (LL-14/LL-15: every measurement needs the null model computed *before* the number is allowed to mean anything) that caught its predecessor, which had confirmed a hypothesis instead.

The completed table: artifact-corrected excess $\sigma - \sigma_{\mathrm{lin}}$ is **positive at all six measurement points** — $+0.040/+1.622/+2.676$ on $z = 0$ and $+0.007/+0.750/+2.073$ on the tilted plane at $\nu = 1/2, 1/10, 1/50$ (the $+0.007$ honestly at noise level), ε-independent, with the exact grid-adequacy stamp passing only at $\nu = 1/2$. The positive control held at literal 0.00 out-of-plane energy *on the tilted plane as well* — the K1 invariance theorem exercised dynamically. The pre-registered kill criterion K3 ($\sigma > 0$ everywhere $\Rightarrow$ the mechanism is dead as a route to regularity for *generic* data) is met at every point.

Verdict (owner-issued, 15 August 2026): OP-2′ is killed for generic data

The fourth translation of the Symmetric-Square lock is dead as a route to global regularity for generic data — and unlike its three predecessors it died *with a number* and with its mechanism understood: the planar manifold is real, exactly invariant, and **repulsive**, increasingly so as viscosity falls. What the verdict does *not* touch: the geometric half (Sym^2 closure ⇒ exact 2D3C confinement, K1-verified in exact arithmetic including tilted planes, apparently unpublished in its lattice form), the torus theorem, and the 5/6 conjecture — all independent results that merely arose during the screening. Four mechanisms have now been proposed and four killed; the programme's rate of *disproof* is its main output to date, and that is what a falsifiable structure is supposed to produce.

11.2 The neighbourhood theorem, scoped by a full read

A first-pass source check reported that the 2D3C manifold has a published global-regularity neighbourhood (Ponce–Racke–Sideris–Titi, Comm. Math. Phys. **159** (1994) 329–341). A full read of the fetched PDF *narrowed* this claim, and the correction is recorded with the same prominence as the claim: the smallness threshold is Gronwall-exponential ($\delta \lesssim \exp\{-C\int_0^\infty \|\nabla v\|^4\}$, eq. (2.17), constants never tracked in ν), hence not quantitatively exploitable against measured σ amplitudes; the 2-D application (their Theorem 4) is pure two-component 2-D on $\mathbb{R}^2$, unforced, with a decay hypothesis periodic solutions violate — *not* 2D3C and *not* the torus; and no periodic-T^3 case is treated. What survives is the concept — a regularity neighbourhood around near-2-D data in adjacent settings, coexisting cleanly with $\sigma > 0$ — not a citable safety net for the Galerkin torus.

11.3 The torus theorem (Tier A) and the 5/6 ball conjecture

While screening OP-2′, the unconstrained resonant-triad 2-section was solved exactly on the torus, and this became the programme's first **Tier A** result about the genuine $\mathbb{Z}^3$-type resonance structure (`TriadTorus.lean`, 7 theorems, clean footprints; derivation memo written and hand-verified first, per LL-5). For any finite additive abelian group G, $\Lambda = G\backslash\{0\}$, counting (triad, slot-pair) *incidences* — the choice that absorbs the $(u, u, 2u)$ degeneracy with no genericity hypothesis:

$$A(u,v) \;=\; 6 - 2\cdot[\,u+v=0\,] \qquad \text{i.e.} \qquad A \;=\; 6(J-I) - 2P,$$

with P the mode-reversal involution. Spectral corollary (as kernel-verified sum identities): degree $6n-8$; zero-sum even vectors are eigenvectors with eigenvalue -8, odd vectors -4; normalised spectrum $\{1,\,-4/(6n-8),\,-8/(6n-8)\}$, so **the torus spectral gap tends to 1**. The measured ball-truncation gap $\approx 5/6$ is therefore a *pure boundary invariant* of the sphere cutoff — an explicitly separated, still-open conjecture (second eigenvalue $1/6$ of the kernel $2\cdot\mathbf{1}_B(x+y) + 4\cdot\mathbf{1}_B(x-y)$ on the unit ball). The measured deviations from 5/6 fall monotonically: 1.7×10^{-3}, 2.4×10^{-4}, 8.6×10^{-5}, 5.9×10^{-5} at $M = 2..5$, 2.5×10^{-5} at $M = 6$ (924 modes, 398,190 triads), and 1.5×10^{-5} at $M = 7$ (1418 modes, 939,930 triads) — seven points, strictly monotone, still **Tier C** evidence for a conjecture, not a result.

12 The Live Band: Formalising Why $\alpha = 1/2$, and What Lies Below

The owner's direction of 15 August 2026 was to target $\alpha \geq 1/2$ as a known theorem — to validate and stabilise — but to *generalise while doing it*, on the view that the real problem is the sign-changing band below $1/2$. Carrying α symbolically through Cheskidov's chain instead of fixing the constant delivers exactly that, and the result is sharper than a formalisation of a known proof would have been.

The threshold is an integrability threshold (Tier A, `DyadicRiccati.lean`)

Write $y = \|u\|^2$. The bilinear estimate $|(B(u,u), Au)| \leq c_b|Au|^{1/\alpha-1}\|u\|^{4-1/\alpha}$ (source; declared sharp there) admits Young absorption against $-\nu|Au|^2$ *iff* $\alpha > 1/3$ — which is the content of the *local* theorem, and **not** where $1/2$ comes from. Absorption leaves a Riccati inequality $y' \leq Cy^s$ with $s = (4\alpha-1)/(3\alpha-1)$, whose blow-up rate is $y \gtrsim (t^* - t)^{-\rho}$, $\rho(\alpha) = 3 - 1/\alpha$. The energy inequality supplies $\|u\|^2 \in L^1_{\mathrm{loc}}$ and nothing more, so a blow-up is refuted precisely when that rate is *not* locally integrable:

$$\neg\,\mathrm{IntegrableOn}\,\left(x \mapsto x^{-\rho(\alpha)}\right)(0,T) \quad\iff\quad \alpha \;\geq\; 1/2.$$

That equivalence is kernel-verified, with the integrability half done in genuine measure theory (Mathlib's characterisation of L^1 for real powers), not as a `Prop`-level placeholder. α enters the entire development at *one* lemma.

Scope, stated in the file and the ledger: this is *not* Cheskidov's Theorem 4.4. It contains no bilinear estimate (an input), no local existence, no Galerkin approximation and no limit passage — which is where the real cost of that theorem sits.

Read backwards, the same theorem quantifies the barrier. Refuting blow-up from a bound $\|u\|^\theta \in L^1_{\mathrm{loc}}$ needs $\theta \geq \theta^*(\alpha) = 2/\rho(\alpha)$, and $\theta^*(\alpha) > 2$ exactly below $\alpha = 1/2$: at $\alpha = 2/5$ one needs $\theta = 4$ and has 2; at $\alpha = 7/20$, $\theta = 14$. The obstruction is a single scalar deficit with exactly two doors — raise θ, or leave the Riccati route — and any future proposal must now name its door and its gain before implementation. That is the pre-registered screen the four dead mechanisms of §10 lacked.

12.1 The screen, and a control broken by its own run

Before searching for a better bound, the cheap question is whether one is even plausible. The probe measures $\int_0^T \|u\|^\theta$ as a function of the shell truncation N: a finite truncation cannot blow up, so a single N proves nothing, and the signature of non-integrability is growth without saturation.

Its theorem-backed positive control failed on the first run — reporting divergence at $\alpha = 1$, where regularity is a published theorem. The cause was neither the integrator nor the observable: the seed $u_n(0) = A2^{-\beta n}$ has finite enstrophy only for $\beta > \alpha$, and at $\alpha = 1$, $\beta = 0.7$ the initial datum has *infinite* enstrophy in the limit. The theorem's hypothesis $u_0 \in V$ was violated, and the measured N-dependence belonged to the initial data rather than the dynamics. The same failure exposed a second defect: runs that stopped early accumulated over *different* time windows, making a diverging control look as though it were saturating. Both are now structural — the harness refuses to run when $\beta \leq \alpha$, and all runs are compared over a common window (LL-17).

After repair the controls behave: $\alpha = 1$ saturates to three decimals, and $\alpha = 1/4$ with large positive data diverges with final times collapsing from 2 to 0.018 — unambiguous blow-up. At moderate amplitude both data classes saturate at $\alpha = 2/5$, but that is the small-data regime where regularity is trivial and saturation carries no information.

The discriminating run, and the third control to save a wrong conclusion

The measurement that matters is at large amplitude, and there the comparison is theorem-backed on one side: Barbato–Morandin–Romito proves positive data at $\alpha = 2/5$ globally regular at *every* amplitude, so that column must saturate however large the data.
It did not. At $A = 8$ and $A = 32$ the positive column showed the same apparent blow-up signature as the sign-changing one — final times collapsing from 2 to 0.08, growth ratios far from 1. Since the positive case is a theorem, the signature had to be an artifact, and it was: the runs were terminated by the integrator's *step-count cap*, not by its magnitude guard. Shrinking final times meant the computation ran out of budget. **The entire amplitude sweep is inadmissible as evidence**, in both columns.
The harness now separates the two stop reasons explicitly and refuses to let a compute-limited block be read at all. This is the third occasion in this campaign where a pre-registered control turned a publishable-looking number into a caught artifact — after the σ null-model catch (§11) and the mis-stated regime band (§13, LL-16). The pattern is consistent enough to be the campaign's main methodological finding: in this domain, the artifact rate for uncontrolled measurements appears to be close to one.

The open question therefore stands exactly where the **Tier A** barrier leaves it, with no numerical evidence in either direction. Rather than build a better integrator, the next step changed the observable and attacked the barrier's first door algebraically, where this programme's instruments are exact rather than approximate.

12.2 Door #1, closed: there is no banded quartic invariant

The barrier needs L^1_{loc} control of $\|u\|^\theta$ with $\theta \geq 4$ at $\alpha = 2/5$. Now $\|u\|^2 = H_\alpha$ is *quadratic* in u and $\|u\|^4 = H_\alpha^2$ is *quartic*, while energy methods produce L^1 control of quadratics — that is what they are. So door #1 reduces to one algebraic question, and it has an exact answer.

Identity (Q1), and what it settles

For $H_\gamma = \sum_n \lambda^{2\gamma n} u_n^2$ on the truncated system,

$$\frac{d}{dt} H_\gamma \;=\; -2\nu H_{\gamma+\alpha} \;+\; 2(\lambda^{2\gamma} - 1) \sum_n \lambda^{(2\gamma+1)n+1}\, u_n^2 u_{n+1}.$$

The nonlinear prefactor vanishes **only** at $\gamma = 0$. Hence *energy is the unique conserved weighted quadratic*: $\theta = 2$ is the entire quadratic supply, not an artifact of technique, and door #1 genuinely requires leaving degree 2. (Q1) generalises this repository's own **Tier A** `shellB_energy_conservation`, which is its $\gamma = 0$ case. A second reading matters as much: for $\gamma < 0$ and *positive* data the prefactor is negative and the sum non-negative, giving a monotone family — which is exactly where positivity does its work in the literature, and exactly what is unavailable for sign-changing data.

An exact rational search for conserved quartics of the inviscid truncated system — diagonal $\sum c_n u_n^4$, neighbour $\sum c_n u_n^2 u_{n+1}^2$, and all banded quartics of index spread below w, up to 75 monomials — returns **nullspace dimension zero in every case**. The pre-registered criterion is met: **door #1 is closed for banded polynomial quartic conserved quantities**, the natural home of an energy-method improvement. Not even E^2 survives, since it is not banded.

What the closure does *not* reach, and what must travel with any citation of it: quartics that are monotone without a polynomial certificate, non-polynomial quantities, quantities conserved only on invariant subsets, and door #2 — leaving the Riccati route altogether. A closed door is not a closed problem.

The search's own control history is worth one line, because it was informative rather than merely corrective. Its first negative control *failed*: perturbing the in-flux exponent $\lambda^n \to \lambda^{n+1}$ does not destroy conservation at all, it merely moves the conserved weights from $c_n = 1$ to $c_n = \lambda^{-n}$. The telescoping is robust to the exponent and sensitive to the *index structure*. That perturbation now serves as a second positive control, and the genuine negative control breaks the index structure instead.

13 What Has Failed, and What That Taught

A programme that reports only successes is not reporting. The following incidents are recorded in `LL.md` with evidence, and each produced a standing process rule.

ID	Incident	Resulting rule
The largest failure to date		
—	**An external expert audit killed this programme's central framing claim** (2026-08-13, verdict D1). For months the specification, this report, and the audit packet all asserted that the formalisation "restates the open core of the Millennium problem". It did not: a 1-D index cutoff, the weight 4^n, and an unconstrained B describe a dyadic shell hierarchy. Four further verdicts (A2, A3, A5, C1) were NO, one (B1) forced the demotion of a whole file from **Tier A** to **Tier C**, and one (B2) required a structural repair. **None of this was caught by two machine gates, five governing documents, or an adversarial self-review** — because none of them can check whether a formal statement refers to the object it names.	**Only human audit certifies statement adequacy, and it must be sought *early* and from *outside*.** A self-review that finds and fixes real gaps (Q3–Q5) creates a false sense of completeness: the gaps it can see are not the structural ones. Corollary now in force: a file that is DRAFT pending audit may not have its informal gloss repeated in prose as though the audit had passed.
Earlier incidents		
LL-1	A broad `git add -A` swept an agent's mid-edit Lean file into a commit; the committed version carried `sorryAx` in five theorems. Happened *twice* in one session.	Stage by explicit path only while any background process may be writing; re-run the gate on the exact file immediately before staging.
LL-2	A submitted proposal self-reported "proven, no `sorry`"; it did not compile (four Lean idiom errors) and its footprint contained `sorryAx`.	Never trust a self-report. Re-run the compiler on the exact artifact. Archive proposals verbatim with their compiled kernel output.
LL-3	D5's redesign fixed stiffness, but the actual obstruction was representation-size growth.	Diagnose the failure mode before redesigning; a fix for the wrong axis is no fix.
LL-4	Two successive designs by the same author failed their feasibility premises.	A repeated failed design is a signal to *escalate the methodology decision*, not to try a third variant alone.
LL-6	A literature claim was nearly cited from memory; the fetched abstract showed the real result was narrower (well-posedness of *positive* solutions in a particular scaling range, not unconditional regularity) [9].	Verify literature precisely before citing, even for a well-known author or result.

ID	Incident	Resulting rule
LL-7	A top-tier-authored design memo's own worked example quoted the wrong summation range.	The certifying agent must trace and correct rather than silently absorb; "derive before dispatch" does not remove the need for downstream re-verification.
LL-8	**The verification gate itself had a false positive.** A lemma proved by purely computational means reports the *strictly cleaner* footprint `[propext]`; the gate string-matched the exact three-axiom list and rejected it as unsound.	Test a changed gate predicate against a table including cases that *must* fail, before trusting a real run; and confirm the script's **exit code**, not its trailing output. The rule's wording in `SPEC.md` was amended to match its intent (membership, not equality).
—	**The Mathlib dependency was never pinned.** The `lakefile` said `require mathlib from git` with no revision; a cold build would have resolved to master and verified everything against a different Mathlib than all existing claims were checked with.	Pin the verifier itself by revision, and track the lockfile. An unpinned prover is a defect in the foundation of a machine-checked programme, not a build-script nuisance.
—	**A proposed correction to the spec would itself have been an error** (the T2 regularity class, §7). The evidence for it was a paraphrased summary, not a primary quotation.	When a correction rests on a summary rather than a direct quote, flag and re-fetch — do not edit the normative rulebook. LL-6 applied to its own application.

14 Summary of Verified State

Artifact	Content	Tier
`CallensDualScale.lean`	T-dual R_{eff} laws, Sym^2 lock, sharpness	**Tier A**
`DyadicShells.lean`	Energy-flux telescoping, monotonicity	**Tier A**
`DyadicShell_Statements.lean`	Shell-model Hypothesis U; concrete `shellB` and its energy conservation (DRAFT)	**Tier A**
`EnstrophyProduction.lean`	Exact enstrophy-production identity	**Tier A**
`EnstrophyProductionBound.lean`	Local bound $S_N^2 \le 2\Omega_N^3$	**Tier A**
`AbstractAlgebraicConservation.lean`	Abstract triad conservation + transversality	**Tier A**
`TriadTorus.lean`	Triad 2-section on a torus: $A = 6(J-I) - 2P$, spectrum	**Tier A**
`DyadicRiccati.lean`	The $\alpha = 1/2$ threshold as an integrability threshold	**Tier A**
Tier A subtotal		
`MillenniumReduction.lean`	Conditional reduction skeleton (DEMOTED 2026-08-13, verdict B1)	**Tier C**
Total kernel-compiled, no axiom outside `{propext, Classical.choice, Quot.sound}`		

All 93 theorems compile with clean axiom footprints; the demoted 7 remain wired into Gate 2 so that they cannot silently rot, but they carry **Tier C** claims and may not be cited as established. The newest row is `TriadTorus.lean` (§11): the torus 2-section solved exactly, with its three negative controls run against scratch perturbed copies before commit (dropping $u \ne v$, dropping no-2-torsion, perturbing -8 to -6: each fails to compile, as required). Eleven Tier B harnesses are wired into Gate 1, each with a negative control demonstrated to fail. All gates pass at commit `b4d0ac0`+, re-run independently for this report.

14.1 What is genuinely established

Exact algebraic identities and inequalities for the truncated dyadic shell model — energy-flux telescoping and monotonicity, the exact enstrophy-production identity, the local production bound, and now the exact energy conservation of the concrete Katz–Pavlović nonlinearity; the T-dual effective radius laws with sharpness; a non-vacuous statement-level formalisation of Hypothesis U *for the shell model*, with nothing abstract remaining in the headline statement, and including a machine-checked refutation of a prior, provably false formulation; and exact-arithmetic verification of two structural properties of the Fourier–Galerkin nonlinearity, one of which is now also proven abstractly.

14.2 What is emphatically not established

Hypothesis U in any form — neither for Navier–Stokes nor for the shell model; any statement about actual Navier–Stokes solutions; any claim that the formalisation reduces, restates, or bears on the Millennium problem (retracted, §2); any claim that the Symmetric-Square Lock is relevant to the NSE cascade; the four tracks (all blocked on undefined objects); any uniformity verdict; the conditional Millennium reduction, whose file is now **Tier C**; and the statement-adequacy of either DRAFT file. The instantiation of B by the *true 3-D* nonlinearity — the 3-D bridge — has not been done and is open.

Definition of done, and what is absent from it

The campaign is complete when the formal skeleton is merged and human-audited, the dyadic results are kernel-verified, the datasets carry a human-issued verdict *whatever that verdict is*, the open problems are either audited or recorded as open with dated escalations, and every ledger row maps to a passing artifact. Note what is *absent*: proving Hypothesis U. If this programme produces a negative dyadic verdict or kills tracks, it **still counts as done and successful as science**. That is what falsifiable structure means — and the audit of 13 August 2026 is the first time that clause has had to be honoured against the programme's own headline framing rather than against one of its subordinate conjectures. It was honoured: the claim was retracted the day it was received, the affected file was demoted, and the programme was re-targeted onto an object its formal artifacts actually describe.

Note on reference verification

Consistent with this programme's own rule LL-6 (*"verify literature precisely before citing, even a well-known author or result"*), every reference below was verified by retrieving the actual arXiv abstract page, publisher listing, or official PDF at the time of writing. **No citation here is written from memory.** Where a detail could not be confirmed by direct retrieval, it is explicitly marked [UNVERIFIED] rather than being asserted. Readers should treat those specific marked details as requiring independent confirmation before reuse.

References

[1] J. Leray. Sur le mouvement d'un liquide visqueux emplissant l'espace. *Acta Mathematica* **63** (1934), 193–248. DOI 10.1007/BF02547354. *Verified* via publisher listing and corroborated by a fetched modern survey (Ożański–Pooley, arXiv:1708.09787), which confirms Leray established local existence of strong solutions, global existence of weak solutions, weak–strong uniqueness, and a singular-set dimension bound. [UNVERIFIED]: the specific attribution of the *mollification* method was not confirmed by direct quotation of the original.

[2] C. L. Fefferman. Existence and smoothness of the Navier–Stokes equation. Official Clay Mathematics Institute Millennium Prize problem description; also in *The Millennium Prize Problems*, CMI/AMS, 2006. *Verified* by fetching the official PDF. The problem asks for a proof of exactly one of four statements: (A) existence and smoothness on $\mathbb{R}^3$; (B) existence and smoothness on $\mathbb{R}^3/\mathbb{Z}^3$ (periodic); (C) breakdown on $\mathbb{R}^3$; (D) breakdown on $\mathbb{R}^3/\mathbb{Z}^3$. https://www.claymath.org/wp-content/uploads/2022/06/navierstokes.pdf

[3] G. Prodi (1959); J. Serrin (1962). The Prodi–Serrin regularity criterion. [UNVERIFIED BY DIRECT FETCH]: the precise inequality, endpoint cases, and constants could not be confirmed from a retrievable full-text source. The criterion as used in this report — a Leray–Hopf weak solution in $L^s_t L^q_x$ with $2/s + 3/q \leq 1$ is smooth — is stated on the authority of secondary sources only. A commonly cited modern treatment is J. C. Robinson, J. L. Rodrigo, W. Sadowski, *The Three-Dimensional Navier–Stokes Equations*, Cambridge University Press, 2016 (*not independently fetched*).

[4] T. Tao. Finite time blowup for an averaged three-dimensional Navier–Stokes equation. arXiv:1402.0290 (submitted 2014; v3 2015). *Verified* by fetching the arXiv abstract page. Constructs an averaged bilinear operator retaining the energy-identity cancellation, and exhibits finite-time blowup for the resulting *averaged* system, via an ODE tied to the Katz–Pavlović dyadic model. **It does not show blowup for the true Navier–Stokes equations**; its force is that regularity proofs must use structure finer than energy methods. [UNVERIFIED]: journal reference not listed on the abstract page.

[5] L. Caffarelli, R. Kohn, L. Nirenberg. Partial regularity of suitable weak solutions of the Navier–Stokes equations. *Comm. Pure Appl. Math.* **35** (1982), 771–831. Pre-dates arXiv. *Verified* via journal citation and corroborating lecture notes: the one-dimensional parabolic Hausdorff measure of the singular set of any suitable weak solution is zero.

[6] T. Buckmaster, V. Vicol. Nonuniqueness of weak solutions to the Navier–Stokes equation. arXiv:1709.10033 (2017); accepted by *Annals of Mathematics*. *Verified* by fetching the arXiv abstract page: weak solutions of 3D Navier–Stokes "are not unique in the class of weak solutions with finite kinetic energy", proved via convex integration. [UNVERIFIED]: exact volume/page numbers.

[7] N. H. Katz, N. Pavlović. Finite time blow-up for a dyadic model of the Euler equations. *Trans. Amer. Math. Soc.* **357**(2) (2005), 695–708; DOI S0002-9947(04)03532-9. *Verified* by fetching and reading the published PDF. Proves finite-time blow-up for the dyadic Euler equations, and for dyadic Navier–Stokes with hyper-dissipation when the dissipation degree is sufficiently small. [UNVERIFIED]: no arXiv posting could be located.

[8] A. Cheskidov. Blow-up in finite time for the dyadic model of the Navier–Stokes equations. arXiv:math/0601074 (2006); *Trans. Amer. Math. Soc.* **360** (2008), 5101–5120. *Verified* by fetching the arXiv abstract page: finite-time blow-up for dissipation degree $\alpha < 1/3$ (improving Katz–Pavlović's $\alpha < 1/4$), and global regularity for $\alpha \geq 1/2$.

[9] D. Barbato, F. Morandin, M. Romito. Smooth solutions for the dyadic model. arXiv:1007.3401 (2010); *Nonlinearity*, DOI 10.1088/0951-7715/24/11/004. *Verified* by fetching the arXiv abstract page. **Scope is narrower than unconditional global regularity**: well-posedness of *positive* solutions of the viscous problem in the scaling range corresponding to Navier–Stokes, and of the inviscid problem in a suitable regularity class. This reference is the subject of lesson LL-6.

[10] J. Bourgain, C. Demeter. The proof of the ℓ^2 decoupling conjecture. arXiv:1403.5335 (2014; v3 2015); *Annals of Mathematics* **182** (2015), 351–389. *Verified* by fetching the arXiv abstract

page. Proves the ℓ^2 decoupling conjecture "for compact hypersurfaces with positive definite second fundamental form and also for the cone".

[11] C. Mouhot, C. Villani. On Landau damping. arXiv:0904.2760 (2009); *Acta Mathematica* **207**(1) (2011), 29–201. *Verified* by fetching the arXiv abstract page, including the journal reference. **The core result requires analytic regularity** (Gevrey-class settings are a related/limiting extension, not the central stated theorem) — see the caution in §7.

[12] F. Golse, L. Saint-Raymond. The Navier–Stokes limit of the Boltzmann equation for bounded collision kernels. *Inventiones Mathematicae* **155** (2004), 81–161. *Verified* indirectly: no arXiv posting located for the 2004 paper, but its existence and scope are confirmed by the fetched follow-up (Golse–Saint-Raymond, arXiv:0808.0039, *J. Math. Pures Appl.* **91** (2009), 508–552), whose abstract explicitly states it completes the earlier *Invent. Math.* 155, 81–161 results. Establishes convergence of renormalised Boltzmann solutions to *Leray* solutions of Navier–Stokes.

[13] H. Duminil-Copin, V. Tassion. A new proof of the sharpness of the phase transition for Bernoulli percolation and the Ising model. arXiv:1502.03050 (2015); *Comm. Math. Phys.* **343** (2016), 725–745. *Verified* by fetching the arXiv abstract page: finiteness of susceptibility below criticality, mean-field bounds above, and subcritical exponential decay of correlations.

[14] A. Giveon, M. Porrati, E. Rabinovici. Target space duality in string theory. arXiv:hep-th/9401139; *Physics Reports* **244** (1994), 77–202. *Verified* by fetching and reading the review PDF. Eq. (2.2.11) establishes invariance of the free-boson partition function under $R/\sqrt{\alpha'} \to \sqrt{\alpha'}/R$ with momentum–winding exchange $m \leftrightarrow n$ — a genuine theorem about the CFT spectrum. [UNVERIFIED]: the literal phrase "self-dual radius" was not found in the pages retrieved, though the fixed-point fact itself is verified. **Important:** the "minimal length" reading is a *physical interpretation* of this exact symmetry, **not a separately proved theorem** — see §3.

[15] C. F. Doran. Picard–Fuchs uniformization: modularity of the mirror map and mirror-moonshine. arXiv:math/9812162. *Verified* by fetching: defines $\mathrm{Sym}^n(L)$ as the operator whose solution space is spanned by products of solutions; Theorem 5.13 states that the Picard–Fuchs equation of a family of M_n-polarized K3 surfaces is the symmetric square of a second-order Fuchsian ODE, hence *third*-order — the standard fact used in §3.

[16] L. de Moura, S. Ullrich. The Lean 4 theorem prover and programming language. *CADE-28* (2021). `https://lean-lang.org/papers/lean4.pdf` *Verified* via converging independent listings (ACM DL, lean-lang.org).

[17] The mathlib Community. The Lean mathematical library. *CPP 2020*, DOI 10.1145/3372885.3373824. *Verified* via converging independent listings.